Convergence of iterated SOCP refinement for CAZAC feasibility

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Abstract

Constant-amplitude zero-autocorrelation (CAZAC) sequences satisfy a constant-modulus condition in time and a flat-spectrum condition in frequency simultaneously, a nonconvex intersection of two tori. Replacing the equality constraints with convex inequalities (bounded amplitude, bounded spectral energy) yields a compact convex semialgebraic body C_τ; at $\tau = 1$, CAZAC systems are exactly its maximal-norm points. The natural algorithm is to iterate a second-order cone program (SOCP) whose objective direction resets to the previous optimizer. This paper's central technical content is Theorem D: a *regularity* hypothesis (R), needed to get a sharp local convergence rate near a CAZAC point, is proved unconditionally at every Zadoff–Chu tuple, for every length n (both parities, not previously closed for even n), via an elementary mechanism new to this problem — the discrete Fourier transform of a Zadoff–Chu chirp is again a chirp, which collapses a linear-dependency rank computation to counting solutions of $t^2 \equiv 0 \bmod n$, so (R) holds iff n is squarefree. We show this closure mechanism is a reusable schema, not a one-off: it extends (with a genuinely different closure fact, the self-duality of the Legendre symbol) to give a partial regularity result at Björck's quadratic-phase construction at prime n — proved, on the symmetry-invariant subspace, and reduced elsewhere to one unproved, numerically-confirmed (44 primes) determinant-nonvanishing lemma — and we prove, by contrast, that the same mechanism provably cannot reach cubic or higher-degree phase sequences. This regularity result underwrites Theorem C (proved conditional on hypothesis (R), unconditional at Zadoff–Chu points of squarefree length): near a regular CAZAC point the ℓ^1 constraint slack has *sharp linear*, not merely quadratic, growth; the KL exponent is exactly 0; and the iteration lands on the point *exactly* after finitely many steps (we give the complete proof of part (c), finite exact arrival). The convergence theory this rests on is Theorem A (unconditional): every measurable selection from the iteration has finite length and converges to a single fixed point, via the Attouch–Bolte–Svaiter Kurdyka–Łojasiewicz (KL) descent framework, applied here to a set-valued argmax update rather than a proximal step; this descent mechanism is itself a known, general-purpose template — Aktaş and Kroer [4] independently instantiate the same KL/Frank-Wolfe argument for a general strongly convex smooth objective, of which our $f = \frac{1}{2}\|x\|^2$ is a special case — and the paper's own contribution here is the reduction of CAZAC feasibility to $\max_{C_\tau} \|x\|^2$ (Prop. 2.2) that makes the template applicable, not the descent argument itself. Since neither that reduction nor the descent argument uses CAZAC-specific structure beyond compactness, convexity, and semialgebraicity, the argument transfers verbatim to a second example, the elliptope $\mathcal{E}_n = \{X \succeq 0 : X_{ii} = 1\}$ — Felzenszwalb–Klivans–Paul's (2021) own worked case, where their finiteness-dependent single-point-convergence mechanism does not apply and their own example stalls at set-level convergence; we prove single-point convergence there unconditionally. Which fixed point is reached is settled only in part: Felzenszwalb–Klivans–Paul's own Theorem 20 already proves, by direct citation and no new argument here, that rank-1 points are exactly the fixed points whose full neighborhood converges to them; a 140-trial numerical sweep finds every trajectory converging to rank 1 with no exceptions; whether the complementary set of initializations has Gaussian measure zero remains open (Q7). Theorem B (unconditional, obstruction): non-CAZAC fixed points exist and some are exposed (unique maximizers of their own functional), so no deterministic selection rule can guarantee convergence to the CAZAC set;

the correct target class of theorem is probabilistic over initializations. We close with a precise statement of the remaining open problems (Q1–Q7) and a tiered verification ledger disclosing exactly which claims are proved, proved conditionally, numerically verified, or conjectural.

1 Setup

Fix $n \geq 1$ and $N \geq 1$. Let F denote the unnormalized discrete Fourier transform on $\mathbb{C}^n$, $F(u)(k) = \sum_m u(m)e^{-2\pi ikm/n}$, so Parseval's identity reads $\|Fu\|^2 = n\|u\|^2$. The state space is $x = (x_0, \ldots, x_{N-1}) \in (\mathbb{C}^n)^N \cong \mathbb{R}^{2Nn}$, with real inner product $\langle u, v\rangle_{\mathbb{R}} = \mathrm{Re}\langle u, v\rangle$ and induced norm $\|x\|^2 = \sum_{j,k} |x_j(k)|^2$. Write the *coupled spectrum* $S_x(k) = \sum_{j=0}^{N-1} |F(x_j)(k)|^2$.

Definition 1.1 (Coupled CAZAC system)**.** x is a coupled CAZAC system if $|x_j(k)| = 1$ for all j, k and $S_x(k) = Nn$ for all k. Write $\mathrm{CAZ}(n, N)$ for this set; for $N = 1$ this is the classical CAZAC condition. $\mathrm{CAZ}(n, N)$ is nonempty for every $n \geq 1$, $N \geq 1$: a Zadoff–Chu sequence exists for every length n [8], and N copies of one satisfy the coupled constraint.

For $\tau \geq 1$, the convex relaxation is

$$C_\tau = \left\{x \in (\mathbb{C}^n)^N : |x_j(k)| \leq \tau \ \forall j, k, \quad S_x(k) \leq Nn \ \forall k\right\}.$$

C_τ is compact (amplitude bounds), convex, semialgebraic, and second-order-cone representable (each constraint is a Euclidean-norm bound on a linear image of x); it is invariant under global phase rotation of each x_j, cyclic shifts, and (for the coupled constraint) any transformation preserving S_x.

Iteration. Given $x^0 \neq 0$, the algorithm is *iterated linear optimization*:

$$x^{\ell+1} \in \arg\max_{x \in C_\tau} \langle x^\ell, x\rangle_{\mathbb{R}}. \tag{$\star$}$$

The argmax may be non-unique; ($\star$) permits any measurable selection. The paper's practical schedule presolves at $\tau = 2 + 1/\ell$ for $\ell = 1, \ldots, 10$, then fixes $\tau = 1$ for the main phase; all theorems below concern the main phase at fixed $\tau \geq 1$ (in particular $\tau = 1$), and hold for any measurable selection rule, so the finitely many presolve steps do not affect any conclusion.

Definition 1.2 (Fixed point)**.** x^* is a fixed point of ($\star$) if $x^* \in \arg\max_{x \in C_\tau} \langle x^*, x\rangle_{\mathbb{R}}$, equivalently $\sigma_{C_\tau}(x^*) = \|x^*\|^2$, where $\sigma_C(r) = \max_{x\in C}\langle r, x\rangle_{\mathbb{R}}$ is the support function of C.

Three increasingly strong convergence questions organize everything that follows: (1) does x^ℓ converge to a single point, not merely have limit points; (2) is that limit CAZAC; (3) at what rate.

2 The iteration as a generalized power method

Proposition 2.1. *With $f(x) = \frac{1}{2}\|x\|^2$ (convex, $\nabla f(x) = x$), ($\star$) is exactly $x^{\ell+1} \in \arg\max_{x\in C_\tau}\langle \nabla f(x^\ell), x\rangle_{\mathbb{R}}$: the conditional-gradient / generalized-power-method update for maximizing the convex function f over the compact convex set C_τ [5], also called iterated linear optimization [3].*

Proof. Immediate from the definitions. □

Proposition 2.2 (Max-norm points are exactly CAZAC points)**.** *For every $x \in C_\tau$ ($\tau \geq 1$), $\|x\|^2 \leq Nn$. For $\tau = 1$, equality holds iff $x \in \mathrm{CAZ}(n, N)$. The maximum is attained for every n, N, τ.*

Proof. By Parseval, $\|x\|^2 = \sum_j \|x_j\|^2 = \frac{1}{n}\sum_k S_x(k) \le \frac{1}{n}\cdot n\cdot Nn = Nn$, with equality iff $S_x(k) = Nn$ for every k. Separately, at $\tau = 1$, $\|x\|^2 = \sum_{j,k} |x_j(k)|^2 \le Nn$ with equality iff every $|x_j(k)| = 1$. Both chains tight simultaneously forces both CAZAC conditions; conversely a CAZAC system attains Nn. Attainment: N copies of a Zadoff–Chu sequence of length n lie in $C_1 \subseteq C_\tau$ with $\text{norm}^2 = Nn$. □

This reframes the nonconvex feasibility problem "find a CAZAC system" as the convex maximization $\max_{x\in C_1} \|x\|^2$, whose optimal value Nn is known a priori: an iterate is δ-suboptimal iff $Nn - \|x^\ell\|^2 = \delta$.

Proposition 2.3 (Monotone ascent, square-summable increments)**.** *Along the iteration at fixed τ, with $x^\ell \neq 0$:*

(1) $\langle x^\ell, x^{\ell+1}\rangle_{\mathbb{R}} \ge \|x^\ell\|^2$;

(2) $\|x^{\ell+1}\| \ge \|x^\ell\|$, *and* $\|x^\ell\| \uparrow \rho \le \sqrt{Nn}$;

(3) $\|x^{\ell+1} - x^\ell\|^2 \le \|x^{\ell+1}\|^2 - \|x^\ell\|^2$, *hence* $\sum_{\ell\ge 0} \|x^{\ell+1} - x^\ell\|^2 \le Nn - \|x^0\|^2$ *and* $\|x^{\ell+1} - x^\ell\| \to 0$.

Proof. (1) $x^\ell \in C_\tau$ is feasible for the SOCP whose optimizer is $x^{\ell+1}$, so the optimum is at least $\langle x^\ell, x^\ell\rangle_{\mathbb{R}} = \|x^\ell\|^2$. (2) Cauchy–Schwarz: $\|x^\ell\|\|x^{\ell+1}\| \ge \langle x^\ell, x^{\ell+1}\rangle_{\mathbb{R}} \ge \|x^\ell\|^2$; divide by $\|x^\ell\|$; monotone and bounded above by Prop. 2.2. (3) $\|x^{\ell+1} - x^\ell\|^2 = \|x^{\ell+1}\|^2 + \|x^\ell\|^2 - 2\langle x^\ell, x^{\ell+1}\rangle_{\mathbb{R}} \le \|x^{\ell+1}\|^2 - \|x^\ell\|^2$ by (1); telescope and apply (2). □

Square-summable increments alone do not imply convergence of the full sequence (harmonic-type sums can wander); closing exactly this gap is Theorem A.

Proposition 2.4 (Limit points are fixed points; CAZAC points are fixed points)**.** *Every limit point of (x^ℓ) is a fixed point of $(\star)$. Every $x^* \in \mathrm{CAZ}(n, N)$ is a fixed point of the $\tau = 1$ iteration, and moreover a global maximizer of its own functional:* $\arg\max_{x\in C_1} \langle x^*, x\rangle_{\mathbb{R}} \ni x^*$ *with optimal value Nn.*

Proof. Let $x^{\ell_j} \to \bar{x}$. By Prop. 2.3(3), $x^{\ell_j+1} \to \bar{x}$ as well. σ_{C_τ} is continuous (finite convex on $\mathbb{R}^{2Nn}$), and $\langle x^{\ell_j}, x^{\ell_j+1}\rangle_{\mathbb{R}} = \sigma_{C_\tau}(x^{\ell_j})$ by definition of the update; pass to the limit. For the second claim, any $y \in C_1$ satisfies $\langle x^*, y\rangle_{\mathbb{R}} \le \|x^*\|\|y\| \le \sqrt{Nn}\sqrt{Nn} = Nn = \|x^*\|^2$ by Cauchy–Schwarz and Prop. 2.2, with equality at $y = x^*$. □

The same Cauchy–Schwarz argument shows any maximal-norm point of *any* compact convex set is a fixed point of iterated linear optimization; nothing CAZAC-specific is used beyond Prop. 2.2 itself — a fact that drives the elliptope corollary in §3.1.

3 Theorem A — global convergence

Theorem 3.1 (Theorem A — Proved, unconditional)**.** *Fix $\tau \ge 1$. For any $x^0 \neq 0$ and any measurable selection $x^{\ell+1} \in \arg\max_{x\in C_\tau} \langle x^\ell, x\rangle_{\mathbb{R}}$, the sequence (x^ℓ) has finite length, $\sum_\ell \|x^{\ell+1} - x^\ell\| < \infty$, and converges to a single fixed point $\bar{x} \in C_\tau$.*

Proof. Apply the abstract descent theorem of Attouch, Bolte, and Svaiter [1, Thm. 2.9] to $F(x) = -\frac{1}{2}\|x\|^2 + \iota_{C_\tau}$ (so minimizing F is maximizing $\|x\|^2/2$ on C_τ):

- **Sufficient decrease** ($a = \frac{1}{2}$): exactly Prop. 2.3(3).

- **Relative error** ($b = 1$): first-order optimality of the argmax step gives $x^\ell \in N_{C_\tau}(x^{\ell+1})$, the normal cone at the new iterate. Since $\partial F(x^{\ell+1}) = -x^{\ell+1} + N_{C_\tau}(x^{\ell+1})$ (sum rule for C^1 plus indicator of a convex set), the vector $w = x^\ell - x^{\ell+1}$ lies in $\partial F(x^{\ell+1})$, and $\|w\| = \|x^{\ell+1} - x^\ell\|$ *exactly*: the subgradient residual is literally the step. This identity is specific to $f = \frac{1}{2}\|x\|^2$; for a generic convex objective the relative-error condition would be a genuine additional hypothesis, but here it is free. (The identical construction, for a general smooth strongly convex objective where it holds only as a Lipschitz-constant inequality rather than an exact equality, is Aktaş–Kroer's [4] Lemma 3.1 eq. (27)–(28); our case is the special instance where that Lipschitz constant is exactly 1, so the inequality becomes an equality — this observation is a corollary of their general lemma, not an independent discovery.)
- **Compactness/continuity**: iterates lie in compact C_τ; F is continuous there.
- **KL property**: C_τ is defined by polynomial (in)equalities and F is polynomial plus the indicator of a semialgebraic set, hence semialgebraic, hence KL [2].

Critical points of F are exactly the fixed points of ($\star$): $0 \in \partial F(x^*)$ iff $x^* \in N_{C_\tau}(x^*)$ (the sum rule above) iff x^* maximizes $\langle x^*, \cdot\rangle_{\mathbb{R}}$ over C_τ, which is the Fixed-point Definition of Section 1. The theorem then gives finite length and convergence to a single critical point. □

The theorem is silent on *which* fixed point the limit is; that is exactly where Theorem B enters. Selection-independence, and the fact that the presolve is finitely many extra solves, mean the conclusion applies verbatim to any concrete implementation of ($\star$).

3.1 A second worked example: the elliptope

The proof of Theorem 3.1 uses nothing about C_τ beyond compactness and convexity, except at the KL step, which needs semialgebraicity. Stated in general: for *any* compact convex $C \subset \mathbb{R}^d$ and the iteration $x^{\ell+1} \in \arg\max_{x\in C}\langle x^\ell, x\rangle$, sufficient decrease (Prop. 2.3(3)'s proof uses only feasibility of x^ℓ for the next problem and Cauchy–Schwarz) and the relative-error identity (the normal-cone argument above, using only $x^\ell \in N_C(x^{\ell+1})$ and $\nabla(\frac{1}{2}\|x\|^2) = x$) hold unconditionally; compactness gives continuity. If C is also semialgebraic, KL holds [2] and Theorem A's conclusion — convergence to a single fixed point, not merely a connected limit set — follows for that C with no CAZAC-specific input.

We exercise this on a genuinely different C: the **elliptope**

$$\mathcal{E}_n = \{X \in \mathbb{R}^{n\times n} : X = X^\top,\ X \succeq 0,\ X_{ii} = 1\ \forall i\},$$

the set of $n\times n$ correlation matrices, with iteration $X^{\ell+1} \in \arg\max_{X\in\mathcal{E}_n}\langle X^\ell, X\rangle$ ($\langle A, B\rangle = \mathrm{tr}(AB)$). This is Felzenszwalb, Klivans, and Paul's own example [3]: the one case in their paper where they concede the fixed-point set is a continuum for $n > 3$, and where their own worked instance stops at set-level convergence.

Proposition 3.2 (Proved)**.** *$\mathcal{E}_n$ is compact, convex, and semialgebraic.*

Proof. Convexity of the constraints is immediate. Compactness: $X \succeq 0$ with $X_{ii} = 1$ gives $|X_{ij}| \le \sqrt{X_{ii}X_{jj}} = 1$ by Cauchy–Schwarz on the associated Gram vectors, so $\mathcal{E}_n$ is bounded, and closed as an intersection of closed sets, hence compact. Semialgebraicity: for symmetric X, $X \succeq 0$ is equivalent by Sylvester's criterion to the finite conjunction of polynomial inequalities "every principal minor of X is ≥ 0"; the diagonal constraints are polynomial equalities. So $\mathcal{E}_n$ is a basic semialgebraic set (a spectrahedron). □

Proposition 3.3 (Proved)**.** *Sufficient decrease, relative error, and compactness/continuity transfer verbatim from Prop. 2.3(3) and the normal-cone argument of Theorem 3.1, replacing "C_τ" by "$\mathcal{E}_n$" and $\langle\cdot,\cdot\rangle_{\mathbb{R}}$ by* $\mathrm{tr}(\cdot\,\cdot)$.

Proof. Neither source proof uses any CAZAC structure; both use only convexity, Cauchy–Schwarz, and the normal-cone characterization of the argmax, all of which hold verbatim on $\mathcal{E}_n \subset \mathbb{R}^{\binom{n+1}{2}}$. $\square$

Proposition 3.4 (Proved)**.** *The fixed-point set of the elliptope iteration is a genuine continuum, not a singleton: $X^* = I_n$ is a fixed point whose optimal face at that direction is all of $\mathcal{E}_n$.*

Proof. For every $X \in \mathcal{E}_n$, $\mathrm{tr}(I_nX) = \sum_i X_{ii} = n$ identically — the linear functional $\langle I_n, \cdot\rangle$ is constant on all of $\mathcal{E}_n$. Hence $\arg\max_{X\in\mathcal{E}_n}\langle I_n, X\rangle = \mathcal{E}_n$ itself: every feasible point, including every rank-1 correlation matrix $vv^\top$ ($v_i = \pm 1$), is simultaneously optimal. This instantiates, with a machine-exact identity, the general fact already noted by Felzenszwalb, Klivans, and Paul [3], who state that $\mathcal{L}_n$ (their name for $\mathcal{E}_n$) has infinitely many fixed points for $n > 3$. $\square$

Proposition 3.5 (Verified numerically, $n = 5$)**.** *Running the iteration on $\mathcal{E}_5$ from four starting points — $X^0 = I_5$; I_5 plus a small random perturbation renormalized to unit diagonal; and two independent random correlation matrices — using an interior-point SDP solver (CLARABEL via cvxpy), all four trajectories show $\|X^{\ell+1} - X^\ell\|_F \to 0$ (to $\sim 10^{-15}$ by iteration 15) and settle on a single matrix. Three of the four escape the fixed-point continuum entirely and converge to a rank-1 correlation matrix $vv^\top$ (eigenvalues $(0, 0, 0, 0, 5)$ to machine precision); the trajectory started at $X^0 = I_5$ is an edge case in which the solver's own deterministic tie-break returns I_5 again, a valid fixed point, not a counterexample. Script:* `experiments/verify_elliptope_convergence.py` *in the companion repository [10]. A broader sweep (same repository,* `experiments/verify_elliptope_rank_sweep.py`*) of 140 random and Gaussian-perturbed trials across $n \in \{3, 4, 5, 8, 10, 15, 20\}$ finds **every trajectory converges to rank** 1, no exceptions. This remains* verified numerically only*, at these tested n and trial counts, not a proof for every n or every trajectory.*

Theorem 3.6 (Elliptope convergence — tiers as stated)**.** *Because $\mathcal{E}_n$ is compact, convex, and semialgebraic (Prop. 3.2), Theorem A's general form applies to it directly: the iterated-linear- optimization sequence on the elliptope converges to a single fixed point for every initialization, **despite the fixed-point set being an infinite continuum** (Prop. 3.4) — **Proved**. Which fixed point is reached is settled only partially: the vertices (rank-1 points) are exactly the fixed points whose full neighborhood converges to them, and no other fixed point has this property, **by direct citation** to [3]'s own Theorem 20 (Remark 3.8 below). That generic nearby trajectories escape the continuum and settle specifically onto a rank-1 point is **verified numerically** (140 trials, Prop. 3.5), not established for every n or every trajectory. Whether the measure of initializations converging elsewhere is exactly zero is **open** (Conjecture, filed as Q7 below).*

Remark 3.7 (Relation to Felzenszwalb–Klivans–Paul). FKP's Theorem 1 (limit points exist, are fixed points, form a connected set) is the same starting point as Prop. 2.4 here, but their only route to single-point convergence is an unnumbered remark that a connected *finite* fixed-point set is a singleton — structurally inapplicable whenever the fixed-point set is a positive-dimensional continuum, which is exactly the situation both here (the $N = 1$ CAZAC phase circle) and in their own §4.2 example. A full read of the paper's body, not merely its abstract, confirms no alternate route exists in their text: the proof of their Theorem 1 only ever derives connectedness and never separately invokes finiteness. Theorem 3.6 closes exactly this gap via the KL/finite-length route [1] rather than the connectedness-plus- finiteness route, and is therefore not subsumed by [3] even in the positive-dimensional-face case. This route itself, as noted at Theorem A, is not unique to this paper — Aktaş–Kroer's [4] independent KL/Frank-Wolfe framework (built on a different but interchangeable abstract descent theorem, Bolte–Sabach–Teboulle–Vaisbourd) would very plausibly deliver the same elliptope conclusion by the same general mechanism; [4] does not discuss FKP, the elliptope, or positive-dimensional critical sets explicitly, so this is not independently confirmed here, but the comparison to FKP should be read as "a route FKP's own paper does not take," not as "a route no other paper could have taken."

Remark 3.8 (FKP's own Theorem 20 bears directly on Q7). Although [3] cannot prove Theorem 3.6's single-point convergence claim, its own Theorem 20 answers a large part of Q7 (below) by direct citation, without any new argument here: *the vertices of $\mathcal{E}_n$ (the rank-1 sign matrices $vv^\top$) are exactly the* ***attractive*** *fixed points of the map $X \mapsto \arg\max_{Y\in\mathcal{E}_n}\langle X, Y\rangle$*, in the pointwise sense that an entire neighborhood of a vertex converges to it, while every non-vertex fixed point admits (their Prop. 19) an explicit curve of nearby points whose norm strictly increases, so it cannot attract a full neighborhood either. This is a genuine dichotomy, not a numerical observation, and it is stronger than what either paper's abstract convergence machinery gives alone: it says which fixed points *can* be limits of nearby trajectories, complementing (not overlapping) Theorem 3.6's unconditional guarantee that *some* single limit exists. It falls short of Q7 as literally stated for a precise reason: [3]'s Prop. 19 produces, in every neighborhood of a non-vertex fixed point, *at least one* nearby point that escapes it — exactly what "not attractive" (their Def. 6) requires — but this does not show *every* nearby point escapes, which is what "repelling" (their Def. 7) would additionally require. Whether non-vertex elliptope fixed points are repelling outright, or can be neither attractive nor repelling (as [3]'s own §3, Example 5, shows can happen in general iterated-linear-optimization systems, for an unrelated three-dimensional cone, not the elliptope itself) is not settled by Theorem 20 either way. If some non-vertex elliptope point is neither, Theorem 20 alone would not rule out a positive-measure set of initializations converging to it through a collective, not individually attractive, mechanism. Q7's residual open content is exactly this measure-zero refinement, structurally the same gap as Q6.

4 Theorem B — obstruction: the limit need not be CAZAC

Theorem 4.1 (Theorem B — Proved, unconditional). (a) *The fixed points of ($\star$) at $\tau = 1$ with every amplitude constraint slack are exactly the points whose Fourier transform has modulus $\sqrt{n}$ on some proper subset $K \subsetneq \{0, \dots, n-1\}$ of frequencies and vanishes off K, with time-domain modulus strictly less than 1 everywhere; such a point has $\|x\|^2 = |K|$.*

(b) *There exist non-CAZAC fixed points that are the* ***unique*** *maximizer of their own linear functional over C_1 — exposed points. For $n = 2$: $x^* = (1, t)$, $t = \sqrt{2} - 1$, is such a point; for $N = 2$, its diagonal doubling $((1, t), (1, t))$ is such a point.*

(c) *Consequently no deterministic convergence-to-CAZAC theorem is possible: every measurable selection rule in ($\star$) retains the point of (b) exactly, once reached.*

Proof of (a), $N = 1$. By Prop. 2.1, $x^* \neq 0$ is a fixed point iff it is a KKT point of $\max_y\langle x^*, y\rangle_{\mathbb{R}}$ subject to $g_m(y) = |y(m)|^2 - 1 \leq 0$ and $h_k(y) = |\widehat{y}(k)|^2 - n \leq 0$; this is a convex program (linear objective, convex constraints) and Slater's condition holds at $y = 0$ ($g_m(0) = -1 < 0$, $h_k(0) = -n < 0$ strictly), so KKT is necessary *and* sufficient for optimality of x^*. The real gradients are $\nabla g_m(y) = 2y(m)e_m$ and $\nabla h_k(y) = 2F^*E_kFy$ ($E_k = e_ke_k^*$), so KKT at x^* reads $x^* = \Lambda x^* + F^*MFx^*$ for diagonal $\Lambda, M \geq 0$, complementary to the active sets. If every amplitude constraint is slack then $\Lambda = 0$, so $x^* = F^*MFx^*$; applying F and using $FF^* = nI$ gives $\widehat{x}^* = nM\widehat{x}^*$ coordinatewise, i.e. for each k either $\widehat{x}^*(k) = 0$ or $\mu_k = 1/n$. Let $K = \{k : \widehat{x}^*(k) \neq 0\} \neq \varnothing$; for $k \in K$

complementary slackness with $\mu_k = 1/n > 0$ forces $|\widehat{x}^*(k)|^2 = n$. Parseval gives $\|x^*\|^2 = |K|$; if $|K| = n$ this forces $\|x^*\|^2 = n$, hence (Prop. 2.2) $x^* \in \mathrm{CAZ}(n,1)$, contradicting amplitude-slackness. So $K \subsetneq \{0,\dots,n-1\}$. Conversely, for such x^* and any $y \in C_1$, $\langle x^*, y\rangle_{\mathbb{R}} = \frac{1}{n}\sum_{k\in K} \mathrm{Re}(\widehat{x}^*(k)\overline{\widehat{y}(k)}) \le \frac{1}{n}\sum_{k\in K}\sqrt{n}\sqrt{n} = |K| = \|x^*\|^2$, attained at $y = x^*$ — a direct support-function argument, no KKT needed for sufficiency. ☐

Proof of (b), $n = 2, N = 1$. For $y = (y_0, y_1) \in \mathbb{C}^2$ feasible in C_1 (so $|y_m| \le 1$ for each m, and $S_y(0) = |F(y)(0)|^2 \le 2$ directly, hence $|F(y)(0)| \le \sqrt{2}$),

$$\langle x^*, y\rangle_{\mathbb{R}} = (1-t)\ \mathrm{Re}\, y(0) + t\ \mathrm{Re}\, F(y)(0) \le (1-t) + t\sqrt{2} = 1 + t^2 = \|x^*\|^2,$$

with equality iff $y(0) = 1$ and $F(y)(0) = \sqrt{2}$, which forces $y(1) = \sqrt{2} - 1 = t$, i.e. $y = x^*$ — a genuine separating functional exposing x^* as a vertex-like point of C_1 even though $x^* \notin \mathrm{CAZ}(2,1)$ ($\|x^*\|^2 = 4 - 2\sqrt{2} < 2$). The $N = 2$ doubling repeats the argument coordinatewise. Part (c) follows immediately from the existence of one such retained point. ☐

Remark 4.2. Theorem A guarantees convergence to *some* fixed point; Theorem B shows the fixed-point set genuinely contains non-CAZAC points that trap every selection rule. The remaining content of "the algorithm works" is a probabilistic/basin-of-attraction statement (Q6 below), not a deterministic one.

5 Theorem C — finite exact termination at regular CAZAC limits

Call $x^* \in \mathrm{CAZ}(n,N)$ **regular** (hypothesis (R)) if the only linear dependency among the gradients of the active constraints at x^* is the one forced by Parseval's identity; equivalently the dependency space K (Section 6 makes this precise) has $\dim K = 1$. Let $\sigma_s > 0$ denote the associated smallest singular value at a regular point.

Lemma 5.1 (Quantitative Newton-type error bound). *Let $\Phi : \mathbb{R}^d \to \mathbb{R}^q$ be a quadratic map (each component a polynomial of degree ≤ 2), $\Phi(x^*) = 0$, $A := D\Phi(x^*)$ surjective with smallest singular value $\sigma > 0$, and let M be a Lipschitz constant for $D\Phi$ (finite and global, since $D\Phi$ is affine). Then for every x with $\|x - x^*\| \le \sigma/(8M)$ and $\|\Phi(x)\| \le \sigma^2/(16M)$,*

$$\mathrm{dist}(x, \Phi^{-1}(0)) \le \frac{2}{\sigma}\|\Phi(x)\|.$$

Proof. Since Φ is quadratic, Taylor's expansion is exact: $\Phi(y+v) = \Phi(y) + D\Phi(y)v + B(v,v)$ with B symmetric bilinear and $\|B(v,v)\| = \frac{1}{2}\|(D\Phi(y+v) - D\Phi(y))v\| \le \frac{M}{2}\|v\|^2$. Run the modified Newton iteration $z_0 = x$, $z_{i+1} = z_i - A^+\Phi(z_i)$, where A^+ is the Moore–Penrose pseudoinverse ($AA^+ = I_q$ since A is surjective, $\|A^+\| = 1/\sigma$). Write $b_i = \|\Phi(z_i)\|$, $d_i = -A^+\Phi(z_i)$, so $\|d_i\| \le b_i/\sigma$. Then

$$\Phi(z_{i+1}) = \Phi(z_i) - D\Phi(z_i)A^+\Phi(z_i) + B(d_i,d_i) = (A - D\Phi(z_i))A^+\Phi(z_i) + B(d_i,d_i),$$

so, using $\|A - D\Phi(z_i)\| \le M\|z_i - x^*\|$,

$$b_{i+1} \le \frac{M\|z_i - x^*\|}{\sigma}\, b_i + \frac{M}{2\sigma^2} b_i^2. \qquad (*)$$

Claim: every z_i satisfies $\|z_i - x^*\| \le \sigma/(4M)$ and $b_i \le b_0 2^{-i}$. By induction: given these, $(*)$ gives $b_{i+1} \le \frac{1}{4}b_i + \frac{1}{32}b_i < \frac{1}{2}b_i$ (using $b_i \le b_0 \le \sigma^2/(16M)$). Summing the geometric increments, $\sum_i \|d_i\| \le (b_0/\sigma)\sum_i 2^{-i} = 2b_0/\sigma \le \sigma/(8M)$, so $\|z_{i+1} - x^*\| \le \|x - x^*\| + 2b_0/\sigma \le \sigma/(4M)$, closing the induction. The z_i are Cauchy with limit z_∞; $b_i \to 0$ and continuity of Φ give $\Phi(z_\infty) = 0$, and $\|z_\infty - x\| \le 2b_0/\sigma$. No convexity is used anywhere in this argument. ☐

Fix $x^* \in \mathrm{CAZ}(n,N)$ and let $H = \{c \in \mathbb{R}^n : \sum_k c_k = 0\}$, P_H the orthogonal projection onto H. Define the **balanced constraint map** $\Phi_s(x) = \big(g(x),\, P_H h(x)/n\big) \in \mathbb{R}^{Nn} \times H$, where $g_{j,m}(x) = |x_j(m)|^2 - 1$ and $h_k(x) = S_x(k) - Nn$ (so $g(x) = 0$ is the amplitude constraint and $h(x) = 0$ is the balanced spectral constraint).

Lemma 5.2 (Zero set). *$\Phi_s^{-1}(0) = \mathrm{CAZ}(n,N)$, globally.*

Proof. If $g(x) = 0$ then x is unimodular, so by Parseval $\sum_k h_k(x) = n\|x\|^2 - nNn = 0$, i.e. $h(x) \in H$; then $P_H h(x) = 0$ forces $h(x) = 0$, so $x \in \mathrm{CAZ}(n,N)$. Conversely $\mathrm{CAZ}(n,N) \subseteq \{g = 0, h = 0\} \subseteq \Phi_s^{-1}(0)$. □

Lemma 5.3 (Uniform derivative bounds). *$D\Phi_s$ is $2\sqrt{2}$-Lipschitz globally, and at any $x^* \in \mathrm{CAZ}(n,N)$, $\|D\Phi_s(x^*)\|_{\mathrm{op}} \le 2\sqrt{N+1}$. Both bounds are independent of n.*

Proof. (Lipschitz.) The rows of Dg are $2x_j(m)$, supported on disjoint real 2-planes, so $\|Dg(x) - Dg(y)\|_F = 2\|x - y\|$. The h_k-row of Dh/n in block j is (the real form of) $2\hat{x}_j(k) f_k/n$ ($f_k(m) = e^{2\pi i k m/n}$), so by Parseval $\|Dh(x)/n - Dh(y)/n\|_F^2 = 4\|x - y\|^2$; P_H is a contraction. Stacking: Lipschitz constant $\le \sqrt{4+4} = 2\sqrt{2}$. *(Operator norm at x^*.)* $Dg(x^*)Dg(x^*)^\top = 4I$ (disjoint unit 2-planes, $|x_j^*(m)| = 1$); the Gram matrix of the unprojected scaled spectral rows is $4N\delta_{kk'}$ (Parseval per block, $S_{x^*} \equiv Nn$). The two blocks have operator norms 2 and $2\sqrt{N}$; the stack has norm $\le \sqrt{4 + 4N} = 2\sqrt{N+1}$. □

Hypothesis (R) restated exactly. $D\Phi_s(x^*)$ is surjective onto $\mathbb{R}^{Nn} \times H$ iff $\dim K = 1$, where K is the dependency space of Section 6: a pair (a, c) with $c \in H$ annihilates the row space of $D\Phi_s(x^*)$ iff $\sum_{j,m} a_{j,m} Dg_{j,m} + \sum_k (c_k/n) Dh_k = 0$, and by the definition of K these solutions are parametrized by $c \in K$ (with a determined by c). The Parseval solution $c = (1, \dots, 1) \notin H$, so the annihilator inside $\mathbb{R}^{Nn} \times H$ has dimension $\dim(K \cap H) = \dim K - 1$ (K always contains the constant vector, which is not in H); hence surjectivity $\iff$ $\dim K = 1$, matching hypothesis (R) as stated above. Write $\sigma_s > 0$ for the smallest singular value of $D\Phi_s(x^*)$ when (R) holds, and $\rho := \sigma_s/(16\sqrt{2})$ (so $\sigma_s/(8M) = \rho$ with $M = 2\sqrt{2}$, Lemma 5.3).

Theorem 5.4 (Theorem C — Proved conditional on hypothesis (R)). *Near a regular $x^* \in \mathrm{CAZ}(n,N)$:*

(a) Sharp (linear) growth: $D(x) \ge \dfrac{\sigma_s}{2\sqrt{2}} \operatorname{dist}(x, \mathrm{CAZ}(n,N))$ *for $x \in C_1$ near x^*, where $D(x) = Nn - \|x\|^2$.*

(b) KL exponent 0: $\operatorname{dist}(0, \partial F(x)) \ge c_0 := \sigma_s/(8\sqrt{2})$ *for all non-CAZAC x in a neighborhood of x^*.*

(c) Finite exact arrival: *if the Theorem A limit $\bar{x}$ is a regular CAZAC point, then $x^\ell = \bar{x}$ **exactly** for all sufficiently large ℓ. The number of "large" steps before exact arrival is at most $(Nn - \|x^0\|^2)/c_0^2$.*

Proof. (a) The objective deficit is simultaneously the ℓ^1 slack of both constraint families: $D(x) = \sum_{j,m}(1 - |x_j(m)|^2) = \frac{1}{n}\sum_k (Nn - S_x(k))$, all terms ≥ 0; in particular $\|\Phi_s(x)\| = \sqrt{\|g(x)\|^2 + \|P_H h(x)/n\|^2} \le \sqrt{D(x)^2 + D(x)^2} = \sqrt{2} D(x)$. *Case 1:* $\sqrt{2}D(x) \le \sigma_s^2/(16M)$, $M = 2\sqrt{2}$. Apply Lemma 5.1 to $\Phi = \Phi_s$ (quadratic; zero set $\mathrm{CAZ}(n,N)$ by Lemma 5.2; surjective at x^* by (R)): $\operatorname{dist}(x, \mathrm{CAZ}(n,N)) \le (2/\sigma_s)\|\Phi_s(x)\| \le (2\sqrt{2}/\sigma_s) D(x)$, i.e. $D(x) \ge (\sigma_s/(2\sqrt{2})) \operatorname{dist}(x, \mathrm{CAZ}(n,N))$. *Case 2:* $\sqrt{2}D(x) > \sigma_s^2/(16M)$, i.e. $D(x) > \sigma_s^2/64$. Since x is in the stated neighborhood, $\operatorname{dist}(x, \mathrm{CAZ}(n,N)) \le \|x - x^*\| \le \rho = \sigma_s/(16\sqrt{2})$, so $(\sigma_s/(2\sqrt{2})) \operatorname{dist}(x, \mathrm{CAZ}(n,N)) \le \sigma_s^2/64 < D(x)$: the bound holds directly. Either case gives (a).

(b) Let $d := \operatorname{dist}(x, \mathrm{CAZ}(n,N)) > 0$ and pick a nearest $z \in \mathrm{CAZ}(n,N)$ (exists, $\mathrm{CAZ}(n,N)$ compact). Any $w \in \partial F(x)$ has the form $w = v - x$ with $v \in N_{C_1}(x)$ (the sum rule, as in Theorem A's proof). Since $z \in C_1$, the normal-cone inequality gives $\langle v, z-x\rangle_{\mathbb{R}} \le 0$, hence, using $\|z\|^2 = Nn$,

$$\langle w, z-x\rangle_{\mathbb{R}} \le -\langle x, z-x\rangle_{\mathbb{R}} = \|x\|^2 - \langle x, z\rangle_{\mathbb{R}} = -\tfrac{1}{2}(Nn - \|x\|^2) + \tfrac{1}{2}d^2 = -(F(x) - F(x^*)) + \tfrac{d^2}{2},$$

so Cauchy–Schwarz on the left gives $\|w\|\, d \ge (F(x) - F(x^*)) - d^2/2$. By (a) (in the form $F(x) - F(x^*) \ge (\sigma_s/(4\sqrt{2}))d$, i.e. (a) with $D(x) = 2(F(x) - F(x^*))$), dividing by d: $\|w\| \ge (F(x) - F(x^*))/d - d/2 \ge \sigma_s/(4\sqrt{2}) - d/2$. Since $d \le \|x - x^*\| \le \rho = \sigma_s/(16\sqrt{2})$, this gives $\|w\| \ge \sigma_s/(4\sqrt{2}) - \sigma_s/(32\sqrt{2}) = (7/32)\sigma_s/\sqrt{2} \ge \sigma_s/(8\sqrt{2}) = c_0$. Taking the infimum over $w \in \partial F(x)$ proves (b).

(c) *(Full chain.)* Let $\bar{x}$ be the Theorem A limit, regular with constants ρ, σ_s, c_0 as in (a)/(b). Since $x^\ell \to \bar{x}$ and $\|x^{\ell+1} - x^\ell\| \to 0$ (Theorem A's square-summability), fix L_0 so that for all $\ell \ge L_0$: $x^{\ell+1}$ lies in the (b)-neighborhood of $\bar{x}$, and $\|x^{\ell+1} - x^\ell\| < c_0$. By the exact identity from Theorem A's proof, $w^{\ell+1} := x^\ell - x^{\ell+1} \in \partial F(x^{\ell+1})$ with $\|w^{\ell+1}\| = \|x^{\ell+1} - x^\ell\|$ *exactly*, at every ℓ – no approximation. Suppose toward contradiction $x^{\ell+1} \notin \mathrm{CAZ}(n,N)$ for some $\ell \ge L_0$. Then $x^{\ell+1}$ is a non-CAZAC point inside the (b)-neighborhood, so (b) gives $\operatorname{dist}(0, \partial F(x^{\ell+1})) \ge c_0$. But $w^{\ell+1} \in \partial F(x^{\ell+1})$ and $\|w^{\ell+1}\| < c_0$ by choice of L_0 — a contradiction, since $w^{\ell+1}$ itself witnesses a subgradient of norm $< c_0$. Hence $x^{\ell+1} \in \mathrm{CAZ}(n,N)$ **exactly**: the identity is exact and (b)'s bound is a hard floor on every non-CAZAC point in the neighborhood, so a subgradient below that floor cannot occur unless the point already lies on $\mathrm{CAZ}(n,N)$. Write $z := x^{\ell+1} \in \mathrm{CAZ}(n,N)$. The next step maximizes $\langle z, y\rangle_{\mathbb{R}}$ over $y \in C_1$; by Cauchy–Schwarz and Prop. 2.2 ($\|z\| = \sqrt{Nn}$ is the global max-norm), $\langle z, y\rangle_{\mathbb{R}} \le \|z\|\|y\| \le Nn$ with equality iff $y = z$ — the argmax is the singleton $\{z\}$, so $x^{\ell+2} = z$, and by induction $x^m = z$ for all $m > \ell$; in particular $\bar{x} = z \in \mathrm{CAZ}(n,N)$. The step-count bound is Theorem A's $\sum_\ell \|x^{\ell+1} - x^\ell\|^2 \le Nn - \|x^0\|^2$ telescoped against c_0^2 per large step (increment $\ge c_0$); since every step after arrival has increment exactly 0 (the sequence is constant at z), every large step occurs strictly before arrival, so this global count already *is* the count of large steps before arrival, with no separate argument needed to exclude post-arrival steps. ☐

Remark 5.5. This is a genuine proof by contradiction, not a heuristic squeeze: an exact identity (equality, from Theorem A) is jointly satisfiable with a hard lower bound (inequality, from (b)) on the same quantity only at dist $= 0$. Parts (a), (b), (c) share one status tag because they rest on exactly the same two inputs — hypothesis (R) at $\bar{x}$ (giving σ_s, ρ) and the exact H2 identity from Theorem A's proof — and (c) adds no assumption beyond the theorem's own hypothesis "$\bar{x} \in \mathrm{CAZ}(n,N)$" plus the unconditional singleton-argmax fact.

Remark 5.6 (The $N = 1$ vs. $N = 2$ gap is local-rate-invariant). Theorem C's constants do not distinguish $N = 1$ from $N = 2$: first-order sharpness at a regular point decides the local geometry identically. The empirically observed difference in success probability between the single-sequence and coupled formulations is therefore a global (basin-of-attraction) phenomenon, not a local-rate one.

Remark 5.7 (Novelty check against generic identifiability theory). Neither the Lewis–Luke–Malick theory of local linear convergence for averaged/alternating projections [6] nor the Bonnans–Shapiro / Drusvyatskiy–Lewis generic identifiability program [7] delivers Theorem C. The former's core hypothesis, strongly regular (transversal) intersection, is exactly what Section 6 shows *fails* at every CAZAC point (the universal Parseval dependency), and even where satisfiable its conclusion is R-linear convergence of a different algorithm, never finite exact arrival. The latter's genericity condition is plausibly satisfied at CAZAC points, but its machinery yields only a qualitative locally-minimal-identifiable-set statement, not a quantitative sharp-growth constant, KL exponent, or finite-step termination guarantee.

6 Theorem D and regularity (R)

Write $x_u(m) = \omega^{-uT_m}$ ($\omega = e^{2\pi i/n}$, $T_m = m(m+1)/2$, n odd) or $x_u(m) = \omega_{2n}^{-um^2}$ ($\omega_{2n} = e^{i\pi/n}$, n even) for the Zadoff–Chu sequence of root u, $\gcd(u,n) = 1$ [8]. At a CAZAC tuple all constraints are active; write

$$K = \{c \in \mathbb{R}^n : \overline{x_j(m)}\,\big(F^*(c \odot \widehat{x}_j)\big)(m) \in \mathbb{R} \ \ \forall j, m\}$$

for the active-gradient dependency space (the tangent space of the constraint intersection has dimension $(N-1)n + \dim K$; hypothesis (R) is $\dim K = 1$).

Theorem 6.1 (Theorem D — Proved for all $n \geq 1$, odd and even)**.** *Let $n = \prod_p p^{a_p}$, $d_{\max} = \prod_p p^{\lfloor a_p/2 \rfloor}$ (the largest d with $d^2 \mid n$). Then for every $n \geq 1$, every $N \geq 1$, every tuple of coprime roots $u_1, \dots, u_N$,*

$$K = \{c \in \mathbb{R}^n : c(k + d_{\max}) = c(k) \ \forall k\}, \qquad \dim_{\mathbb{R}} K = d_{\max},$$

independent of N, of the roots, and of the parity of n.

Proof. **Step 0.** A dependency $\sum_{j,m} a_{j,m} \nabla g_{j,m} + \sum_k c_k \nabla h_k = 0$ evaluated on a perturbation is equivalent to $F^*(c \odot \widehat{x}_j)$ being coordinatewise a real multiple of x_j — the defining condition of K; dependencies with $c = 0$ force $a = 0$ since amplitude gradients have disjoint supports. So the dependency space is (isomorphic to) K.

Step 1 (self-transform). For odd n, writing $k = us$, $\widehat{x}_u(k) = G_u\, \omega^{uT_{u^{-1}k}}$ with $G_u = \sum_j \omega^{-uT_j}$, $|G_u|^2 = n$ by Parseval, using the exact identity $T_{m+s} = T_m + T_s + ms$. For even n, the parallel identity holds with T_m replaced by m^2, exponents tracked mod $2n$: $\widehat{x}_u(k) = G_u\, \omega_{2n}^{us^2}$, using $\omega_{2n}^{v(j+n)^2} = \omega_{2n}^{vj^2}$ (an exact integer identity valid because n is even).

Step 2 (reduction to a correlation equation). Substituting Step 1 into the dependency condition and reindexing $k = us \bmod n$ turns $c \in K$ into reality, for all m, of a convolution $\sum_s c'_s\, e(s+m)$, where e is built from G_u and the same phase.

Step 3 (Fourier solve). Taking the length-n DFT in m reduces this to $\hat{c}'(t) = 0$ wherever $\hat{e}(-t) \neq 0$. In both parities, $\hat{e}(t) = 0$ exactly when $\sigma^2 \equiv 0 \pmod{n}$ ($\sigma = u^{-1}t$): the zero set is the subgroup $Z_0 = \{t : t^2 \equiv 0 \bmod n\} = m_0\mathbb{Z}_n$, $|Z_0| = n/m_0 = d_{\max}$ where $m_0 = \prod_p p^{\lceil a_p/2 \rceil}$.

Step 4. $\hat{c}'(t) = \hat{c}(u^{-1}t)$, so "$\operatorname{supp} \hat{c}' \subseteq Z_0$" is the same condition for every root and block, giving $K = \{c : \operatorname{supp} \hat{c} \subseteq m_0\mathbb{Z}_n\}$, i.e. the $d_{\max}$-periodic real vectors, of real dimension $d_{\max}$. This count is read off directly from $|Z_0| = d_{\max}$ and is insensitive to parity: an alternate derivation via counting conjugate-symmetric pairs $\{t, -t \bmod n\}$ in Z_0 would need to treat the self-paired frequency $t = n/2$ separately when $n \equiv 0 \bmod 4$ places $n/2 \in Z_0$, but that bookkeeping is specific to the pair-counting route and does not arise here, since Steps 1–4 characterize K by its Fourier support directly. ☐

Remark 6.2 (Scope of the written derivation). Steps 2–3 compress a Gauss-sum completion-of-square computation of the same kind as Step 1's self-transform identity (the correlation $c' \star e$ of Step 2 and its Fourier zero-set in Step 3 both reduce, by the same technique, to evaluating quadratic exponential sums); the symbol-by-symbol expansion is not spelled out here, in the same spirit as Lemma 6.7 below, whose "exact symbolic computation" is likewise not displayed in full. The *conclusion* $\dim K = d_{\max}$ is independently corroborated: by hand at $n = 4$ (self-paired case) and $n = 9$ (odd, non-self-paired) via direct enumeration of Z_0, and numerically across all $21 + 43 = 64$ tested Zadoff–Chu configurations (both parities; Section 6's corollaries below cite the exact scripts). No counterexample or numerical discrepancy has been found at any tested configuration.

Corollary 6.3. *Hypothesis (R) holds at every Zadoff–Chu tuple iff n is squarefree — in particular at every prime n, odd or $n = 2$. Theorem C is therefore* ***unconditional*** *whenever the Theorem A limit lies on the symmetry orbit of a Zadoff–Chu tuple and n is squarefree.*

Corollary 6.4. *At Zadoff–Chu tuples with squarefree n (either parity), the CAZAC tangent dimension is exactly $(N-1)n+1$: a 1-dimensional phase circle for $N = 1$ versus an $(n+1)$-dimensional manifold for $N = 2$.*

Numerical verification: `experiments/verify_zc_regularity.py` for odd $n \in \{3, 5, 7, 9, 15, 21, 25, 27, 45, 49, 75\}$ (21 configurations) and `experiments/verify_even_n_regularity.py` for the 13 even values $n \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 24, 36, 50\}$ (43 configurations); all match $\dim K = d_{\max}$ exactly. No Chebotarëv-type nonvanishing-minors argument is needed: the DFT of a chirp is a chirp, collapsing the dependency analysis to counting solutions of $t^2 \equiv 0 \bmod n$. Both scripts are in the companion repository [10].

6.1 The chirp-argument schema, and where it does and does not generalize

Stripped of Zadoff–Chu-specific notation, Theorem D's proof needs exactly: (1) a closure property, $\widehat{x_u} = G_u \cdot x_u \circ \phi_u$ for a scalar G_u and an index-set automorphism ϕ_u; (2) an algebraic addition law for the phase making the key substitution exact; (3) applying the closure twice, once to reduce the dependency condition to a correlation equation, once more to diagonalize that equation into a congruence on the Fourier support of the kernel. This replaces a linear-algebra rank question with a congruence-counting question, and is a genuinely reusable schema, not a one-off computation.

Proposition 6.5 (Positive lead — Proved classically, newly connected here)**.** *The Legendre symbol χ modulo an odd prime p satisfies $\hat{\chi}(k) = \chi(k)\, g(\chi)$ for $k \neq 0$ ($g(\chi)$ the Gauss sum, $|g(\chi)|^2 = p$): χ is, up to a scalar, its own DFT, the same closure shape used by Theorem D, with trivial reindexing.*

Proposition 6.6 (Negative result — Proved, standard exponential-sum theory, newly connected here)**.** *Cubic and higher-degree phase ("algebraic chirp") sequences do not have this closure property: quadratic Gauss sums have an exact closed-form evaluation (magnitude $\sqrt{n}$ always), which is what underwrites Theorem D's Step 1, but Weyl sums of degree ≥ 3 have no such exact evaluation, only upper bounds, and the addition law $(m+s)^d = m^d + s^d +$ (nonlinear mixing) for $d \geq 3$ does not reduce to a reindexing of the original phase.* ***The specific self-transform closure mechanism used in Theorem D's proof — an exact, modulus-$\sqrt{n}$ evaluation of the underlying exponential sum — does not extend to cubic or higher phase sequences****; this rules out the route used here, though it does not itself preclude some structurally different argument reaching the same conclusion by other means.*

6.2 The Björck case: a partial result, not a proof

Björck's construction (odd prime p, Legendre symbol χ) [9] defines $b(m) = e^{i\theta(m)}$ with θ a two-valued function of $\chi(m)$, differing by $p \bmod 4$. Unlike χ itself, b is not literally self-dual under the DFT, but a different closure fact rescues part of the argument.

Lemma 6.7 (Proved, exact symbolic computation)**.** *b is exactly* ***affine*** *in χ: $b = A \cdot \mathbf{1} + B \cdot \chi + C \cdot \delta_0$ with $C = 1 - A$, for explicit constants A, B depending on $p \bmod 4$. The 3-dimensional space $T := \mathrm{span}_{\mathbb{C}}\{\mathbf{1}, \chi, \delta_0\}$ closes under the DFT: $\hat{b} = C \cdot \mathbf{1} + Bg \cdot \chi + Ap \cdot \delta_0$.*

Lemma 6.8 (Proved)**.** *Let G be the group of quadratic-residue units mod p (order $q = (p-1)/2$), acting by decimation. $b, \hat{b}$ are pointwise G-invariant, so K is a G-invariant subspace of $\mathbb{R}^p$. $\mathbb{R}^p \cong \mathbf{1}_{\{0\}} \oplus \mathbb{R}[G] \oplus \mathbb{R}[G]$ as a G-module, with trivial-isotypic piece exactly $T \cap \mathbb{R}^p$ (dimension 3).*

Theorem 6.9 (Proved, both residue classes)**.** $K \cap T = \mathrm{span}\{\mathbf{1}\}$*; i.e. restricted to the* 3*-dimensional symmetry-invariant candidate space, the only dependency is the Parseval one, for every prime* p*.*

Proof. Restricting the dependency condition $\overline{b(m)}\,(F^*(c\odot\hat{b}))(m) \in \mathbb{R}$ for all m to $c = \alpha\mathbf{1}+\beta\chi+\gamma\delta_0 \in T$ collapses — since $b, \hat{b} \in T$ closes under pointwise product and F — to exactly three scalar equations, one evaluation each at $m = 0$, $m \in \mathrm{QR}$, $m \in \mathrm{QNR}$ (the condition depends on m only through $\chi(m) \in \{0, 1, -1\}$). Write α_0 (resp. β_0) for the phase such that $Bg = s\sqrt{p}\,e^{i\alpha_0}$ with $s = \pm 1$ when $p \equiv 1 \pmod 4$ (resp. the corresponding phase when $p \equiv 3 \pmod 4$), using that the Gauss sum $g = g(\chi)$ is real for $p \equiv 1 \pmod 4$ and purely imaginary ($g = i\sqrt{p}$, up to sign) for $p \equiv 3 \pmod 4$ — this parity of g is exactly where the two residue classes diverge. The three equations reduce, after eliminating the $m = 0$ equation (which involves only α, γ and is solved by α free), to:

$$p \equiv 1 \pmod 4: \quad s\sqrt{p}\,\beta = 0, \qquad -s\gamma\big(1+c(p-1)\big) = 0 \quad (c = \cos\alpha_0 = \pm 1,\ s = \pm 1 \text{ the sign of } Bg);$$

$$p \equiv 3 \pmod 4: \quad \frac{\sqrt{p}(1-\cos\beta_0)}{2}\,\beta = \frac{p\sin\beta_0}{2}\,\gamma, \qquad \frac{\sqrt{p}(1-\cos\beta_0)}{2}\,\beta = -\frac{\sin\beta_0}{2}\,\gamma.$$

In the first case, $s \neq 0$ so $s\sqrt{p} \neq 0$ forces $\beta = 0$ directly. Since Bg is real (g is real for $p \equiv 1 \pmod 4$), $\alpha_0 \in \{0, \pi\}$ exactly, so $c = \cos\alpha_0 = \pm 1$ exactly (not merely bounded); then $1 + c(p-1)$ equals p (if $c = 1$) or $2 - p$ (if $c = -1$), both nonzero for $p \geq 5$, so $1 + c(p-1) \neq 0$ forces $\gamma = 0$. In the second case, subtracting the two equations gives $\gamma\big(\frac{p\sin\beta_0}{2} + \frac{\sin\beta_0}{2}\big) = 0$, i.e. $\gamma \sin\beta_0 (p+1)/2 = 0$; since $p + 1 \neq 0$ and $\sin\beta_0 \neq 0$ (a degenerate $\beta_0 \in \{0, \pi\}$ would force g real, contradicting $g = i\sqrt{p}$), $\gamma = 0$, and then $\beta = 0$ from either equation (the coefficient $\sqrt{p}(1 - \cos\beta_0)/2$ is nonzero unless $\beta_0 = 0$, again excluded). In both residue classes the system forces $\beta = \gamma = 0$ with α free, giving $K \cap T = \mathrm{span}\{\mathbf{1}\}$ exactly. □

Remark 6.10 (What is proved and what is not — do not upgrade). Lemma 6.7 and Lemma 6.8 and Theorem 6.9 are complete proofs. What remains **unproved** is whether K meets any of the $q - 1$ nontrivial isotypic pieces of $\mathbb{R}^p$ under G: by Schur's lemma the dependency operator restricted to each such (complex 2-dimensional) block is a 2×2 matrix, structurally a Jacobi-sum-type bilinear expression, and its nonvanishing was not proved. A second, independent attempt at a clean closed-form nonvanishing argument also failed: the relevant determinants are confirmed numerically nonzero at every one of 44 tested primes $p \in [5, 200)$ (both residue classes, $\dim K = 1$ throughout), but with no uniform lower bound of the shape a classical single-Gauss/Jacobi-sum-modulus argument would give — at $p = 71$ the relevant gap is a factor of $\approx 4 \times 10^3$ smaller than at the next-nearest tested prime, a genuine near-degeneracy; the script itself reports this was independently confirmed not to be a floating-point artifact by a separate 60-digit-precision recomputation at that specific prime (performed once, outside the script's own run, and recorded in its output). Script: `experiments/verify_bjorck_isotypic.py` in the companion repository [10] (the 44-prime determinant check; a separately-run set of 14 primes, all contained in this same tested range, is used for the affine-closure check of Lemma 6.7, in `experiments/verify_bjorck_regularity.py`). **The overall regularity claim at Björck points is therefore still open, not proved**, despite this genuine partial progress: what is established is a full proof restricted to the symmetry-invariant subspace, plus strong (but non-uniform) numerical evidence, not a proof, for the complementary directions.

7 Open problems

Q1 (settled). Does x^ℓ converge to a single point? Yes — Theorem 3.1.

Q2 (open, reframed). Characterize all fixed points of ($\star$) for general n, N. The stationarity system is a KKT eigenvalue equation $x^* = (\Lambda + F^*MF)x^*$ for diagonal multipliers, structurally reminiscent of discrete prolate/Slepian time-and-band-limiting operators. Complete for $n = 2$ (four symmetry classes); open in general. A sharper sub-question: are non-CAZAC fixed points always measure-zero repellers, or can they attract an open set of initializations?

Q3 (partially answered). Why does $N = 2$ empirically always succeed where $N = 1$ sometimes fails? The dimension lower bound $\dim \mathrm{CAZ}(n, N) \geq (N-1)n + 1$ (proved) gives a mechanism, and Corollary 6.4 makes it exact at squarefree Zadoff–Chu points; but the pathological fixed-point classes are the same for $N = 1, 2$, so the gap is about basin measure, not existence, and remains open.

Q4 (essentially settled locally; one remaining gap). Rate of convergence near $\mathrm{CAZ}(n, N)$: settled by Theorem C at regular points, and Theorem D now makes regularity unconditional at Zadoff–Chu points of either parity for squarefree n. Remaining: (R) at non-Zadoff–Chu CAZAC points is open beyond the Björck partial result above; n-uniformity of the true sharp-growth constant is conjectural (proved bounds scale like c_N/n, sampled ratios show no decay).

Q5 (open, deprioritized). What does the τ-continuation presolve actually buy? Hypothesis: it drives the iterate onto (or near) the easier flat-spectrum variety before the main phase. Untested by ablation.

Q6 (the central open question). For $N \geq 2$ and Gaussian initialization, does the iteration converge to $\mathrm{CAZ}(n, N)$ with probability 1 (or $\geq 1 - c^n$)? Requires a full fixed-point census (Q2) and a basin/measure argument. For $N = 1$, empirical success probability decays from ≈ 1 at small n to ≈ 0.3 at $n = 167$; whether this decay is exponential or polynomial in n is unresolved.

Q7 (partially settled by citation, residual part open). Does every trajectory of the elliptope iteration converge to a rank-1 point? Theorem 3.6 proves single-point convergence unconditionally but not convergence to an extreme point specifically. The *pointwise* half of this question is already settled — Remark 3.8 cites [3]'s own Theorem 20: rank-1 points are exactly the fixed points whose full neighborhood converges to them, and every other fixed point admits an explicit escaping direction. What remains open is the *measure-theoretic* refinement: does the set of initializations whose trajectory converges to a non-vertex fixed point have Gaussian measure zero? [3]'s own escaping-curve argument (Prop. 19, Remark 3.8) only produces one escaping point per neighborhood, not universal escape, so it leaves open whether non-vertex elliptope points are repelling outright or could instead be collectively (not individually) attracting, the way [3]'s unrelated cone example (§3, Example 5) shows can happen in general for this style of iteration. This residual question is structurally the same conjecture as Q6 in a second, unrelated domain: if the measure is zero, it is independent evidence for Q6's general shape; if not, CAZAC's extra structure (the dimension count of Corollary 6.4) may be doing more work than currently credited. The numerical evidence (Prop. 3.5, 140 trials across $n \in \{3, \ldots, 20\}$) shows zero exceptions, consistent with — but not a proof of — the measure-zero reading. *Attack plan:* a basin/measure argument analogous to Q6's, using the pointwise dichotomy of Theorem 20 as the starting classification of which points can even be candidates for a positive-measure limit.

8 Verification status and provenance

Tiers. *Unconditional*: complete proof, no open hypothesis. *Proved conditional on (R)*: complete proof, resting on regularity hypothesis (R). *Proved conditional on (R) and an open sub-case*: complete proof modulo one explicitly identified, unresolved lemma. *Verified numerically*: no proof; a script reproduces the claim to a stated tolerance. *Conjecture*: stated as a target, not established either way.

Principal claims.

- Propositions 2.1–2.4 (generalized power method, max-norm = CAZAC, monotone ascent, limit points fixed): *Unconditional*.
- Theorem A (global single-point convergence): *Unconditional* as a mathematical claim, but its descent-argument mechanism is a special case of Aktaş–Kroer's [4] general Frank-Wolfe/KL framework (2025); the paper's own contribution here is the CAZAC-to-max $\|x\|^2$ reduction (Prop. 2.2) that makes that template apply, not the descent argument itself — see the note at Theorem A.
- Elliptope semialgebraicity and H1–H3 transfer (Props. 3.2–3.3): *Unconditional*. Fixed-point continuum (Prop. 3.4): *Unconditional* (cited fact plus a machine-exact identity). Single-point convergence despite that continuum (Theorem 3.6): *Unconditional* — this is a proved theorem, not a numerical finding. The pointwise vertex/non-vertex attractive dichotomy (Remark 3.8): *Unconditional*, by direct citation to [3] Theorem 20 (not a new argument here). *Separately*, that generic trajectories escape onto a rank-1 point specifically (Prop. 3.5): *Verified numerically only* (140 trials, $n \in \{3, \dots, 20\}$) — this is not proved. The residual, measure-zero form of universal rank-1 convergence (Q7): *Conjecture*.
- Theorem B (obstruction, exposed non-CAZAC fixed point): *Unconditional*.
- Theorem C, parts (a),(b),(c), including the full contradiction-based proof of part (c): *Proved conditional on hypothesis (R)*; all three parts carry the same status tag.
- Theorem D (regularity of Zadoff–Chu points, $\dim K = d_{\max}$), **both odd and even** n: *Unconditional*, a complete proof for all n regardless of parity, though Steps 2–3 of the written proof are a compressed Gauss-sum computation rather than a symbol-by-symbol expansion (see the "Scope of the written derivation" remark after the proof) — independently corroborated by hand at $n = 4, 9$ and numerically at all 64 tested configurations, but a reader wanting the full symbolic derivation must reconstruct Steps 2–3 by analogy with Step 1, as that remark discloses. Corollaries 6.3–6.4: *Unconditional* (Corollary 1 conditionally activates Theorem C, but is itself unconditional as a dimension/regularity statement).
- Legendre-symbol closure positive lead: classical fact, newly connected to the schema, *Unconditional* as stated (no claim of Björck regularity is made from it alone). Cubic/higher-phase negative result: *Unconditional* (a genuine proved negative, not a placeholder).
- Björck partial result: Lemma 6.7, Lemma 6.8, Theorem 6.9 ($K \cap T = \mathrm{span}\{\mathbf{1}\}$): *Unconditional*. Full regularity (R) at Björck points: **open**, reduced to one unproved determinant-nonvanishing lemma, *verified numerically only* at 44 primes, $p \in [5, 200)$. Do not read this as "proved" — the reduction is real but incomplete.
- Q1–Q7: precise open-problem statements as in Section 7; Q1 is settled (restated as a theorem above), the rest range from "partially answered" to fully open conjectures, as stated there.

Provenance. This document was written with substantial assistance from large language models (Claude, Anthropic) under the direction of Jeffery Kline, drawing on a longer working record of derivations kept in the companion repository [10]'s `theory/` directory; every numerical claim above is backed by a named script in that repository's `experiments/` directory. Every claim above carries the status tag most consistent with that record; none has been upgraded in the writing of this paper.

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[10] J. Kline, *cazac-algorithm*: companion research repository holding the `theory/` derivations, the dated `AUDIT-LEDGER.md`, and the `experiments/` scripts (including `verify_zc_regularity.py`, `verify_even_n_regularity.py`, `verify_elliptope_convergence.py`, `verify_bjorck_regularity.py`, `verify_bjorck_isotypic.py`) that reproduce every numerical claim in this paper, 2026.